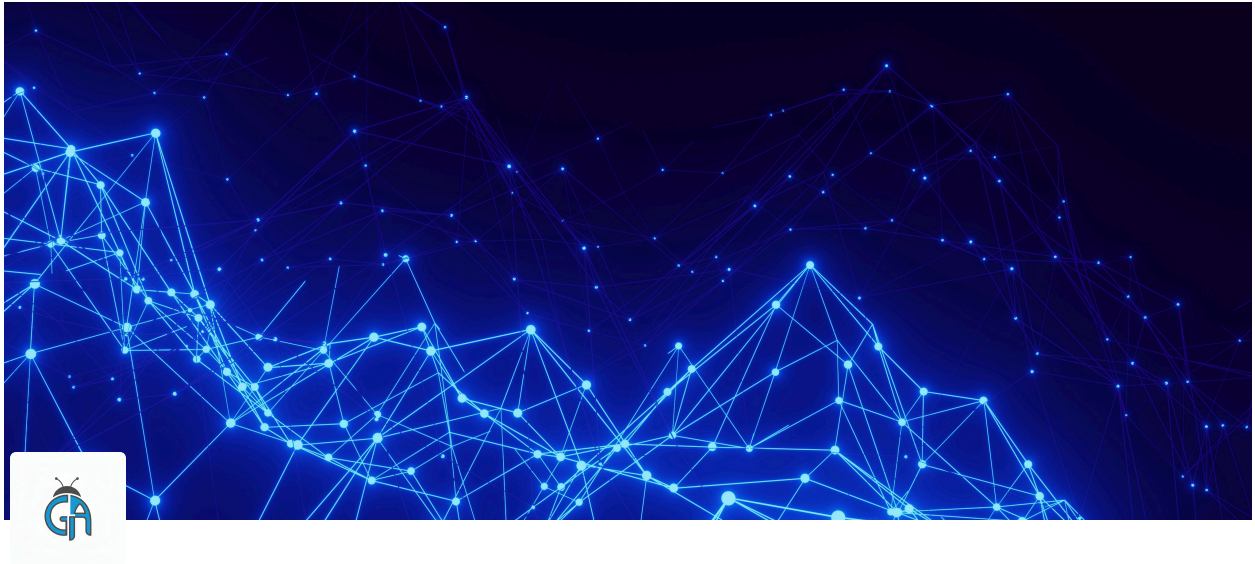




TDD vs. BDD

Coding vs. Conversation

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TL;DR

TDD (Test-Driven Development): Developers write tests for the *code* (Unit Tests) before writing the code.

BDD (Behavior-Driven Development): The team writes tests for the *behavior* (User Scenarios) in plain English before writing the code.

The Core Difference

Both approaches write the test **first** (Test-First approach), but the audience is different.

TDD (The Developer's Tool)

- **Focus:** "Am I building the code correctly?"
- **Language:** Code (C#, Java, Python).
- **Process:** Red ➡ Green ➡ Refactor.

1.

- a. Write a failing test (Red).
- b. Write just enough code to pass it (Green).
- c. Clean up the code (Refactor).

BDD (The Team's Tool)

- **Focus:** "Am I building the right feature?"
- **Language:** Plain English (Gherkin: Given/When/Then).
- **Process:** Conversation ➡ Specification ➡ Automation.

2.

- a. PO, Dev, and QA discuss the requirement.
- b. They write a Scenario together.
- c. Dev implements the code to make that Scenario pass.

The Construction Analogy

TDD (The Brick):

The engineer tests the strength of a single steel beam before installing it.

"Does this beam hold 500kg?"

(Internal Quality).

BDD (The Blueprint):

The architect and the homeowner agree on the layout of the kitchen.

"When I stand at the sink, I want to see the garden."

(External Behavior).

The Comparison Matrix

Component: Audience

- TDD: Developers (Technical).
- BDD: Everyone (Dev, QA, Product Owner).

Component: Syntax

- TDD: Programming Language (e.g., `Assert.AreEqual(2, result)`).
- BDD: Human Language (e.g., `Given I am logged in`).

Component: Main Benefit

- TDD: Better code design, fewer bugs in logic.
 - BDD: Better communication, living documentation.
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Real World Example: A Login Feature

TDD Test (Written in C#):

```
[Test]
public void TestPasswordEncryption() {
    var result = Encrypt("password123");
    Assert.AreNotEqual("password123", result);
}
```

Checks if the internal encryption function works.

BDD Scenario (Written in Gherkin):

```
Scenario: Successful Login
Given I am on the Login Page
```

When I enter valid credentials
Then I should be redirected to the Dashboard

Checks if the user can actually use the feature.

Why This Matters to Me

As a QA moving into Automation, **BDD** is my bridge.

I can write test scenarios in **Gherkin** (which I already know from Manual Testing), and then use tools like **SpecFlow** or **Cucumber** to turn those English sentences into automated tests.